Lẩu gà Lá É
Software Development Plan (Small Project)
Version <2.0>

Revision History

Date	Version	Description	Author
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30/10/2026	1.5	Refinement on Project Organization	Nguyễn Hoàng Gia
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Software Development Plan (Small Project)

1. Introduction

1.1 Purpose

The purpose of this *Software Development Plan* is to detail all the necessary information in order to make an efficient plan for developing the product. Thís included defining clear scope for the product, setting out reasonable objectives to achieve, and aligning multiple stakeholder visions of said product.

This document benefits not only the lead of the project - by facilitating a plan and easing the need for monitoring - but also individual members, by providing more transparency about the process and mitigating risk of misalignment.

1.2 Scope

This *Software Development Plan* details the plan for <project-name>, which includes but is not limited to the development process, the order of development, key objectives of the <project-name>, how the work is divided, what is the quality control process, etc.

1.3 Overview

This *Software Development Plan* contains the following information:

Project Overview — detailing the goal of the project, what it set out to achieve, as well as the scope and limitation of said project. It will also clarify what will be the expected result of this development, and what assumptions will be made in said development.

Project Organization — describes the organizational structure of the project team, which include information such as the role of each member, the responsibility of each member and team and the expected workload during the development.

Project Plan — provides the comprehensive plan for the development process, details such as the estimated schedule, task assignments table and build plan will be provided in this section.

2. Project Overview

2.1 Project Purpose, Scope, and Objectives

Our project is a single-vendor e-commerce web application for an electronics store, aimed mainly at customers who are not familiar with technical specifications or only have a basic level of tech literacy. We aim to develop a software solution for the task of online shopping, not only just to simplify the process, but also to be a helpful assistant during that process.

By the end of this project, we aim to deploy a fully functional web page, where customers can browse and search products, compare specifications, add items to a cart, check out, pay, and track their orders, and where store staff can manage products, stock, orders, promotions, and reviews. A key objective is our AI integration into the system, that recommends and compares products from everyday-language requests, helps with cart and checkout preparation, and answers order and stock questions, while the customer always confirms the order and payment.

The system is a desktop web application for modern browsers (Chrome and Firefox), requires an internet connection, and delegates card and e-wallet payments to an external payment gateway.

2.2 Assumptions and Constraints

2.2.1 Assumptions

The following assumptions are used to establish the initial project plan. These assumptions will be reviewed throughout the development process and updated if circumstances change.

Assumption	Description
Team Member Availability	All five team members are assumed to be available to participate in development, testing, documentation, and Scrum activities according to the agreed project schedule.
Development Environment	Each team member is assumed to have access to a computer, an internet connection, and the necessary development tools to work on their assigned components.
Product Data Availability	Sufficient electronics product information, including product names, brands, prices, images, and technical specifications, is assumed to be available for building the product catalog, comparison features, and recommendation system.
API Integration	The Backend, Frontend, and AI components are assumed to be able to communicate through agreed API contracts. Required APIs are expected to become available according to the integration schedule.
AI Service Availability	The team assumes that suitable AI models or external LLM services will be accessible within the project's technical and financial limitations.
Testing Environment Availability	The team assumes that local development environments will support initial testing and that a remote testing environment can be made available before the planned integration and deployment tests.

2.2.2 Constraints

The project must operate within the following academic, resource, technical, and organizational constraints.

Constraint	Description
Academic Schedule	The project must follow the university's assignment schedule. Development and documentation must be organized using Scrum, with the final Sprint supporting a demonstration of the completed application.
Team Size and Resources	The development team consists of five students who share responsibilities across Backend, Frontend, AI, testing, documentation, and project management. The project scope must remain achievable with the available workforce and time.
Technology Stack	The project uses React and Vite for the Frontend, Java Spring Boot and PostgreSQL for the Backend, and Python FastAPI for AI services. Significant technology changes require team agreement.

System Architecture	The project is organized into three separate GitHub repositories for Frontend, Backend, and AI. The Backend is responsible for business logic, authentication, authorization, and business data management. The AI service communicates with the Backend through APIs rather than directly accessing the business database.
Platform Requirements	The application is developed as a web-based e-commerce platform targeting desktop users. It must support modern browsers, particularly Google Chrome and Mozilla Firefox, on the operating systems identified in the Project Proposal.
Security Requirements	The system must protect sensitive customer information and credentials. Authentication and authorization must be enforced for protected operations. AI-assisted shopping actions must respect user permissions, and final order submission must require explicit customer confirmation.
Development and Collaboration Standards	The team must follow the agreed Git workflow, including feature branches, Pull Requests, peer review, and testing before merging changes. Jira is used to manage tasks, assignments, and project progress.
Documentation and Submission Requirements	Project documentation must be written in English using Markdown. Required diagrams must use Mermaid, and submission documents must include their Markdown and PDF versions. Each document section must identify its performer, reviewer, and editor.

2.2.3 Pending Decisions

Some project conditions have not yet been finalized and may affect the assumptions and constraints above.

Pending Decision	Potential Impact
Deployment Platform and Budget	The team must confirm whether to use AWS or another hosting platform and determine the available infrastructure budget.

These decisions will be incorporated into the project plan once they have been reviewed and approved by the team.

2.3 Project Deliverables

The team will deliver software builds and project documents throughout five Sprints. Exact university submission dates follow the official course schedule. Jira Sprint boundaries are used for planning, not treated as confirmed submission deadlines.

Sprint	Deliverables	Target date
Sprint 1 (PA1)	Project Proposal, Team Contract, Existing Apps Survey, and initial architecture.	PA1 submission; exact date not recorded.
Sprint 2 (PA2)	Project Plan, Vision Document, Spec Kit artifacts, AI Usage Report, Weekly Report, and initial Backend, Frontend and AI foundations.	PA2 deadline TBD; Jira Sprint 2 ends 30 Oct 2026.
Sprint 3 (PA3)	Updated documents after TA feedback, authentication, catalog/search, cart, and basic Frontend–Backend integration.	Sprint 3 / PA3: TBD.
Sprint 4 (PA4)	Checkout, payment, order management, administration and integrated AI functions; a testing build.	Sprint 4 / PA4: TBD.
Sprint 5 (PA5)	Complete web application, test evidence, deployment instructions and final demonstration.	Sprint 5 / PA5: TBD.

Four testing builds are planned for Sprints 2–5. Their scope and acceptance criteria are described in Section 4.2.3 Releases.

3. Project Organization

3.1 Organizational Structure

The team has five members. Gia is the Project Manager and coordinates the Sprint plan, assignments and integration work. Development is organized into Backend, Frontend and AI workstreams. Team members may contribute to more than one workstream.
Backend — Lead: Trịnh Văn Thành; Member: Nguyễn Hoàng Gia.
Frontend — Lead: Nguyễn Công Kiên; Members: Lê Đức Thịnh and Phan Thị Thanh Huyền.
AI — Lead: Lê Đức Thịnh; Member: Nguyễn Hoàng Gia.
Technical leads coordinate development and review changes in their areas. All members participate in Scrum meetings, documentation, self-testing and cross-testing.

3.2 Roles and Responsibilities

Person	Role	Main responsibilities
Nguyễn Hoàng Gia	Project Manager; Backend and AI member	Manage project progress, Sprint tasks and documentation; develop product catalog/data tools and the recommendation system.
Trịnh Văn Thành	Backend Lead	Lead Backend APIs, authentication, authorization, database integration,

		CI/CD and deployment.
Nguyễn Công Kiên	Frontend Lead	Lead Frontend architecture, routing, customer interfaces and Backend API integration.
Lê Đức Thịnh	AI Lead; Frontend member	Develop the AI Shopping Agent, LLM integration and AI–Backend contracts; support chatbox/UI work.
Phan Thị Thanh Huyền	Frontend member	Develop homepage and administrator interfaces; support UI testing, documentation and reporting.

Each Jira task has one primary owner and a different teammate as reviewer. Team members report blockers early and help test integrated features.

4. Management Process

4.1 Project Estimates

Given the limited timeline of this course, we have made the following estimation related to the timeline and resource:

- Software size: Our proposal includes 10 function groups with about 33 features in total (2–5 per group). The AI assistant, checkout and payment, and the admin side are expected to be the largest parts.
- Effort: Works will be divided among 5 members (see Project Organization section). Assuming a fair division of task, it is estimated that each member needs to devote about 7.5 hours per sprint, or close to 40 hours throughout 5 the whole project. In total, roughly 200 hours, and this covers development, testing, documentation and meetings.
- Time: The whole project' timeline is controlled entirely by the course, so the date of our final build is expected to be delivered 11 weeks after the start of this project, or 9 weeks since the first proposal was sent.
- Cost: Development costs are kept as low as possible.
    - Backend and API: During development the backend runs locally on team members' machines, at no hosting cost. Once the core features are stable, we plan to deploy it, most likely on AWS. The deployment cost is still [TBD] and will be estimated once the platform is confirmed.
    - LLM API**:** The AI assistant uses an external LLM API with a budget of at most 1 USD per month.
    - Email service**:** Password-reset and restock emails use a free tier, or a console/log output in development.
    - Payment**:** Payments are tested through the gateway's sandbox at no cost.

4.2 Project Plan

4.2.1 Phase Plan

Each Sprint has one Planning meeting, two Scrum meetings and one Review. Every task has one owner and a different reviewer.

The team will keep the current Jira Sprint 2 dates: 22–30 October 2026. This Sprint is shorter than the course recommendation of 2–3 weeks. Issue due dates still need to be added.

The team plans to complete all features in the PA1 Project Proposal by PA5. Progress and workload will be reviewed at the end of every Sprint.

Sprint 2 focuses on PA2 documentation, Backend APIs and product data, Frontend pages, and AI foundations.

Table 1. Five-Sprint roadmap and planned deliverables

Sprint	Objective	Main deliverables
1 (PA1)	Project initiation and architecture	Project Proposal, Team Contract, app survey, architecture
2 (PA2)	Development foundations and documentation	Project Plan, Vision, Spec Kit, reports; core API/UI/data/AI foundation
3 (PA3)	Core shopping functionality	Authentication, catalog/search, cart and FE–BE integration
4 (PA4)	Extended commerce and AI integration	Checkout, order/payment, admin, recommendation and agent integration
5 (PA5)	Complete and demonstrate PA1 scope	All committed features, regression results, final demo

4.2.2 Iteration Objectives

Sprint goals are shown in Table 1. Tasks for later Sprints will be assigned during Sprint Planning.

Table 2. Sprint 2 work packages (current Jira ownership)

Workstream	Owner(s)	Sprint 2 Jira issue(s) / outputs
Backend — platform	Thành	SWE-39,40,42,44,46: authentication, users, media, CI/CD, engagement
Backend — product data	Gia	SWE-41,43,45,113,114: catalog, normalize, import, setup, remote import
Frontend — customer	Kiên	SWE-47–52: routes, access, API client, product and auth
Frontend — admin/UI	Huyền, Thịnh	SWE-53–61,116–119: layouts, design, pages, chatbox
AI services	Gia, Thịnh	SWE-62–69: recommendation and shopping agent foundations
PA2 documentation	Team (per Jira)	SWE-70–112: Project Plan, Vision, Spec Kit, reporting

Integration checks: Gia and Kiên test the Catalog API with Frontend pages; Gia and Thịnh agree on AI data and API contracts; Thành manages CI/CD.
A task is Done only after implementation, self-testing and required review. Incomplete tasks move to the next Sprint and remain visible in Jira.

4.2.3 Releases

The team plans four testing builds (Table 3). Thành leads CI/CD and deployment. Build dates after Sprint 2 are still TBD.

Table 3. Planned testing builds and release points

Build	When	Scope	Acceptance checkpoint
B1 Foundation	Sprint 2; Jira window ends 30 Oct 2026	Catalog/data, service startup, FE–BE foundations	Basic build, API smoke and integration checks; release unverified
B2 Core Shopping	Sprint 3; date TBD	Authentication, catalog, search/filter, cart	Component, auth, API/integration and shopping regression tests
B3 Commerce + AI	Sprint 4; date TBD	Checkout, payment, orders, admin, recommendation, agent	Business state, permission, failure and integration checks
B4 Final Demo	Sprint 5; date TBD	All PA1 scope and final demo	E2E, regression, security checks, defect retesting

A build is accepted when it starts correctly, passes its planned tests and has recorded results. Critical security, payment, stock or order errors block acceptance.

Table 4. Testing levels and primary responsibilities

Testing activity	Primary responsibility
Unit / component	Each implementing developer
Backend API / contract	Gia, Thành; FE/AI owners support
Frontend / UI	Kiên, Huyền, Thịnh
AI recommendation / tools	Gia, Thịnh
Cross-service integration	Relevant Backend, Frontend, AI owners
End-to-end and regression	All five members
Preliminary performance / deployment	Gia, Thành; CI/CD owner: Thành

Local machines are used for development and basic tests. The remote server is for deployment, integration, preliminary performance checks and the final demo when ready.
For each build, record the tested version, test results, defects and retest outcomes. Track bugs in Jira.
Quality checks: Frontend — npm run lint, npm run build; AI — Ruff and pytest; Backend — Maven verification. Actual test results must be saved.

4.2.4 Project Schedule

Milestones follow five Sprints (PA1–PA5). The current Jira Sprint 2 runs from 22 to 30 October 2026. Dates for Sprints 1, 3, 4 and 5 are not confirmed in the available schedule. The team reviews progress and deliverables at the end of each Sprint. Individual task due dates must be added to Jira and aligned with the final Project Plan.

Sprint	Schedule	Milestone
Sprint 1 (PA1)	Date not recorded	Project Proposal, Team Contract, initial architecture
Sprint 2 (PA2)	22–30 Oct 2026 (Jira)	Project Plan, Vision, Spec Kit, foundational code
Sprint 3 (PA3)	TBD	Core shopping features and FE–BE integration
Sprint 4 (PA4)	TBD	Checkout, payment, orders and AI integration
Sprint 5 (PA5)	TBD	Full-system tests and final demonstration

4.2.5 Project Resourcing

This is an undergrad course project, and therefore the number of staff/members is fixed (5), and it is expected to not be changed throughout the course.

In our group, the 5 members are grouped into 3 different domains: backend, frontend and AI development (2 members of the AI team also work in a different domain).

- Backend team [2 members]: both has have prior experience with backend framework implementation and API gateway
- Frontend team [3 members]: All three have some familiarity with HTML and CSS, and two have limited UI design skills. The project is mainly in React, in which all three have limited experience.
- AI team has [2 members]: both have worked together in developing an AI product in the past and integrated it into one of the prior course's applications.

4.3 Project Monitoring and Control

4.3.1 Requirements Management

Requirements come from the Project Proposal and Vision Document. New requests are recorded in Jira, reviewed by the technical lead and approved by the team before work starts.
Changes to shared APIs must be communicated to Backend, Frontend and AI owners. Approved changes are reflected in Jira and documentation.

4.3.2 Reporting and Measurement

Jira workflow: To Do → In Progress → Review → Done. Each issue should have an owner, reviewer, expected deliverable and due date. Members update their own issues.

The team holds one Planning meeting, two Scrum meetings and one Review per Sprint. Gia monitors overall progress. Messenger is used for quick updates, Discord for meetings, and When2Meet for scheduling.

Table 5. Progress measures and reporting cadence

Measure	Review frequency	Evidence
Task status / ownership / due dates	Ongoing; Scrum meetings	Jira board and issue history
Blockers and carryover	As discovered; Sprint Review	Jira blockers, Sprint Review notes
PR review and test checks	Every Pull Request	GitHub PR and CI logs
Integration and build readiness	At build checkpoints	Smoke/API/E2E evidence, build records
Documentation and training	Each PA submission	Markdown/PDF, reviewer metadata, screenshots/logs

When a task is blocked, the owner informs the team. The lead helps resolve it; Gia reviews the schedule impact. Unfinished work moves to the next Sprint and Jira is updated.

4.3.3 Risk Management

Project risk is evaluated at least once per iteration and documented in this table. The risks of the greatest magnitude are listed first in the table.

Risk Ranking	**Risk Description and Impact**	**Mitigation Strategy and/or Contingency Plan**
High	AI assistant features may be too ambitious within the given timeframe. Inability to develop such technical features would result in a failure at finishing the project proposal.	The core features (browsing, checkout, reviewing, etc.) should be developed first, and the AI feature will be developed in parallel by a separate team. A fallback would be to deploy a reduced assistant, with the ability to answer basic questions.
Medium	Schedule slippage, where any of the members of the group fail to meet the deadline that has been set out at the start of the sprint. Missed time accumulated may cause a great delay in the project schedule	The lead of each group will be regularly checking in members of their respective group. This is to ensure that the work at hand can be accomplished within the given timeframe. If there are any cases where the task may not be finished in time, it must be communicated as soon as possible to find a remediation.
Medium	Frontend and backend integrations problem. API may be changed mid project, or different assumptions are made by the 2 separated teams. Miscommunication would result in the webpage crashing or failing to meet expectations.	Agree on the API contract early, and have the frontend use mock data until the backend is ready. Announce any API change in the group chat and update the spec. Build a communication channel between the team, as well as weekly meetings to discuss progress and roadblock.
Medium	Dependence on third-party services (email, server, LLM API) through downtime, rate limits,	Use sandbox modes, wrap each service behind one module so it can be swapped, and mock it in development. Set a budget or usage cap for the LLM

	cost or sandbox access.	API.
Low	A member dropped out or suffered a long duration of inaction (due to external issues). Their work is stalled and code knowledge is disrupted.	Clear documentation in every step for a smooth transfer if necessary. Keep each task small, and cross review each other's PR. Assign a backup owner for some specific domain.
Low	Late or insufficient testing, with bugs found just before the demo.	Test each build when it completes, and make “app runs and lint passes” part of the PR checklist. Reserve time before each milestone for fixes only.

4.3.4 Configuration Management

The team has four GitHub repositories: Frontend-SWE, Backend-SWE, docs-SWE and AI-SWE. Work is done on a separate branch and merged through a reviewed Pull Request. Do not push directly to main. Branch examples: feature/catalog-api, fix/login-validation, docs/project-plan. Commit example: feat: add catalog API. PRs should link to Jira and include test evidence.

Documents use English Markdown, Mermaid diagrams where needed, and PDF for submission. Each section names its performer, reviewer and editor.

Do not commit passwords, keys or .env files. Backend controls authentication and business data; AI accesses it through authorized APIs.

If a build fails, the team checks logs, fixes or reverts the change, and tests again. Backup details are added when the server setup is finalized.